

BY 539: Programming and applied AI/ML for life sciences

Course Code	BY 539
Course Name	Programming and applied AI/ML for life sciences
L-T-P-C	2-1-0-3
Prerequisites	None
Intended for	PG & PhD
Mutual Exclusion	IC 272, ME 630, MB 522, and CS 672
Approval	65th BoA (Aug 2026)

Preamble

The rapid growth of high-throughput technologies has generated large and complex datasets in the biological and life sciences domains, including genomics, bioimaging, physiology, and neuroscience. Machine Learning and artificial intelligence (AI) methods are increasingly important for analysing such datasets, identifying patterns, predicting biological outcomes, and supporting data-driven research. This course is specifically tailored for students majoring in Biotechnology and Biology, with little or no prior background in programming, data science, or computational methods. It provides a gradual, biology-focused introduction to programming, data analysis, machine learning, and AI, using familiar biological and biomedical examples and datasets to develop computational skills without requiring an extensive computer science background.

The course is primarily intended for students of Biotechnology and Biology and is also open to M.Tech. and PhD scholars from other schools/centres whose research work is in the life sciences or involves biological/biomedical applications, subject to the instructor's permission. Scholars from other disciplines may enrol if their research has a substantial life-science component and they have the required academic background.

The course emphasises tutorial/practical, hands-on learning with real-world biological and biomedical datasets, enabling students to develop computational skills that can be applied directly to areas of life sciences such as bioinformatics, biomedical signal analysis, neuroscience, disease classification, and biological data modelling.

Course Modules

- **Unit/Topic 1: Programming:** Introduction to programming, data types and structures, control statements, functions, basic object-oriented programming, shell scripting including basic Unix/Linux commands, variables, input/output, conditional statements, loops, functions, file handling, command-line arguments, and scientific and numerical computing. (9 Hours)
- **Unit/Topic 2: Data Analysis and Visualization:** Introduction to datasets and data types, data handling, data cleaning and transformation, exploratory data analysis, descriptive statistics, correlation, data visualisation and basic statistical analysis. (6 Hours)
- **Unit/Topic 3: Introduction to Machine Learning:** Fundamentals of artificial intelligence and machine learning, supervised and unsupervised learning, data preprocessing, feature engineering, regression, classification, clustering, dimensionality reduction, and model evaluation and applications to time-series data. (10 Hours)
- **Unit/Topic 4: Fundamentals of Deep Learning:** Artificial neural networks, Introduction to Perceptron & MLP, activation and loss functions, forward and backpropagation, optimisation and regularisation, convolutional neural networks (CNNs), recurrent neural networks (RNNs/LSTMs), and applications to standard datasets such as Physio NET datasets, UniProt (8 Hours)

Project: yes

Laboratory Modules (9 hours)

Students will receive hands-on training in programming, data preprocessing and visualisation, statistical analysis, and implementation of machine-learning and deep-learning models to real-world and biomedical datasets.

Text Books

1. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (3rd ed.). O'Reilly Media.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

References

1. Raschka, S., Liu, Y., & Mirjalili, V. (2022). *Machine Learning with PyTorch and Scikit-Learn*.
2. Yang, X.-S. (2014). *Nature-Inspired Optimization Algorithms*. Elsevier.